

OAL-TA-001: Temporal Aliasing Incorporation Framework

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Document Purpose

This document captures the first formal incorporation of **Temporal Aliasing** into the OAL research and simulation architecture. It builds upon and integrates with three prior OAL framework documents: **Conditional Failure at the Root Level (CFTRL)**, the **Open Window Doctrine (OWD)**, and the **Presentation-State Integrity Framework (PSIF)**.

The central thesis is that Temporal Aliasing — originally a signal processing concept describing the corruption of data when a system samples a signal too slowly — is also a fundamental property of biological cognition, multi-device technological ecosystems, and simulated environments. This document defines the concept, establishes its cross-domain relevance, proposes a Simulation Lab framework for testing it, and outlines a real-world implementation roadmap.

This document serves as the connective tissue between the three prior OAL frameworks, formalizing temporal aliasing as the shared underlying mechanism that CFTRL addresses at the root level, OWD addresses at the interface-transfer level,

and PSIF addresses at the presentation-vs.-ground-truth level. Temporal Aliasing Incorporation brings these threads into a unified architectural statement.

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Closing Notice

1. Executive Summary

- **Temporal aliasing is a universal architectural constraint, not a technical edge case.** Any system that samples a continuous reality at

discrete intervals — including electronic signal processors, numerical simulations, autonomous vehicle sensor arrays, and the human brain — is subject to temporal aliasing. The human brain operates on intrinsic alpha-band perceptual "frames" of approximately 10–13 Hz, producing a structural sampling limitation at the biological root layer of all observation and decision-making.

- **Human perception is reconstructed, not recorded.** The brain does not render an accurate portrayal of the information it processes. What an individual perceives as objective reality is a best-effort reconstruction — an amalgamation derived from incomplete, asynchronous, multi-modal sensory streams and prior-knowledge interpolation. Predictive coding research confirms that expected inputs are suppressed and mismatches are amplified, meaning the version of events that reaches conscious awareness is systematically biased before any external distortion is introduced. Every human operates from this reconstructed version as if it were ground truth, because no higher epistemological access point exists.
- **Multi-device ecosystems create a structural temporal aliasing cascade.** Each technology device — smartphone, smartwatch, smart television, laptop, voice assistant — operates on its own independent clock, refresh rate, rendering pipeline, audio latency, and network timestamp. A typical household or office environment presents 5–10 such devices simultaneously, each offering a competing "version of now." The biological system attempting to reconcile these streams operates at a fundamentally different sampling frequency than any of the devices. The resulting errors in temporal reconstruction are not malfunctions; they are the predictable, structural output of a profound sampling mismatch.
- **The three prior OAL frameworks each implicitly address aspects of temporal aliasing.** CFTRL addresses temporal aliasing at the simulation-stack root level — the silent failure mode in which the root simulation produces aliased output that no nested tier can detect. OWD addresses it at the interface and cross-environment transfer level — the mechanism by

which artifacts timed to fall within the brain's temporal binding window bypass detection. PSIF addresses it at the presentation-vs.-ground-truth level — the structural difficulty of provenance-based reconstruction when the observer's own experiential log is aliased before any external deception begins. Temporal Aliasing Incorporation formalizes the connecting tissue between all three.

- **A testable, gate-governed Simulation Lab framework is herein proposed.** The OAL Temporal Aliasing Lab (TA-Lab) operates across three tiers: Biological (human psychophysics and temporal binding window measurement), Technological (instrumented multi-device environment with ground-truth logging), and Simulational (controlled time-step aliasing injection in numerical simulation environments). A four-gate progression protocol governs advancement, with ONESTAMP required at Gate 2 and Gate 4 transitions. A structured funding pathway from seed funding through federal grant scale is mapped to each gate.
- **This document introduces the concept of Temporal Provenance Corruption as a named OAL architectural term.** When temporal aliasing corrupts the observer's own internal record of when events occurred, provenance-based reconstruction — the audit method proposed by PSIF — is working from a distorted log. Combined with Presentational Inertia (the environment resists acknowledging the gap between presentation and state), Temporal Provenance Corruption creates a double-locking mechanism: the environment resists correction, and the observer's capacity to perceive the gap is itself structurally compromised. Formal treatment of this compound effect is a priority item for near-term OAL development.

3. Concept Overview — What Is Temporal Aliasing?

3.1 Signal Processing Definition

Temporal aliasing occurs when a continuous signal is sampled at a rate below the **Nyquist-Shannon threshold** — that is, at a rate less than twice the highest frequency component of the signal being sampled. When this condition is violated, high-frequency content in the original signal is not captured accurately; instead, it "folds back" into the lower-frequency range of the sampled data, appearing as a spurious lower-frequency artifact. The resulting aliased signal is structurally indistinguishable from genuine lower-frequency signal to any downstream observer or processor. This corruption is *irreversible*: once the aliased data is recorded, the original continuous signal cannot be recovered from that record alone.

The governing mathematical relationship for the aliased perceived frequency is:

$$f_{\text{perceived}} = |f_{\text{signal}} - N \times f_{\text{sample}}|$$

where N is the nearest integer such that the result falls within the range $[0, f_{\text{sample}}/2]$. The classic empirical illustration is the **"wagon wheel effect"** in film and video: a wheel spinning at 170 Hz, sampled at a camera frame rate of 200 Hz, is perceived to rotate at only 30 Hz — and depending on the exact relationship between signal and sample frequency, the apparent direction of rotation may be reversed.

Crucially, the aliased frequency is not noise. It is *structurally false signal that appears indistinguishable from true signal* to any observer operating downstream of the sampling event. This property — the structural falseness that is undetectable from within the sampled record — is the defining characteristic that makes temporal aliasing architecturally dangerous in layered or nested systems.

3.2 The Nyquist Constraint as a Universal Principle

The Nyquist-Shannon sampling theorem is typically understood as a constraint specific to electronic signal processing — analog-to-digital conversion, audio engineering, image capture. However, its logical structure applies universally: **any observer sampling a continuous process at discrete intervals is subject to aliasing** when its sampling rate is insufficient relative to the highest frequency in the signal it is attempting to capture. This applies to cameras, microphones, EEG electrodes, lidar sensors in autonomous vehicles, numerical simulation time-steppers, and — critically for the OAL framework — to the human brain's own perceptual architecture.

A 2024 paper by Arruda and Moreno (CENTAI Institute / University of Zaragoza) formally established that Nyquist-Shannon limits apply to numerical simulations of complex systems. Their analysis demonstrated that time steps that are too large introduce spurious frequencies into simulation output that make the simulation appear to model a physical reality that does not exist. The implication is stark: a poorly-sampled simulation is not merely inaccurate — it is *generating hallucinated dynamics* that are internally self-consistent but disconnected from the actual system being modeled. This finding directly informs the OAL TA-Lab's Tier 3 (Simulational) protocol and the CFTRL root-level silent failure scenario [1].

3.3 Temporal Aliasing in the Brain — The Intrinsic Frame Rate Hypothesis

The "continuous wagon wheel illusion" — a version of the motion-reversal effect that occurs even under continuous, non-stroboscopic illumination — provided early evidence that the visual cortex itself operates on a discrete sampling rhythm rather than as a continuous processor. The leading hypothesis is that this rhythm operates near the **alpha band (10-13 Hz)**, creating an intrinsic "frame rate" of approximately 10-13 perceptual snapshots per second in the human visual cortex.

This hypothesis has received formal mathematical support from Parks and Kilpatrick (University of Colorado Boulder, 2026), whose work using delay-coupled neural field models demonstrates that neural propagation delays generate coexisting traveling bump solutions with distinct, quantized propagation speeds. The brain's internal perception of motion is organized into *discrete dynamical states*, not a smooth continuous stream. Incoming continuous reality is being discretely sampled, and the resulting perception is a reconstruction — one capable of producing reversed motion percepts, time compression, and spurious state transitions in precisely the same way that electronic temporal aliasing does [2].

This is not a peripheral quirk of visual processing. It is a structural property of the neural architecture underlying all temporal perception. The brain is a temporal aliasing system operating at a biologically determined frame rate, receiving continuous reality, and producing a reconstructed, quantized approximation as its output — which it then presents to consciousness as unmediated reality.

3.4 Related Temporal Phenomena — Comparative Table

Phenomenon	Domain	Core Mechanism	Effect on Observer	OAL Relevance
Temporal Aliasing	Signal Processing / Neuroscience	Undersampling of high-frequency signal below Nyquist threshold	False lower-frequency artifact perceived as real; structurally undetectable from within sampled record	Root mechanism for all subsequent OAL temporal framework entries; parent concept
Wagon Wheel Effect (Visual)	Visual Cortex / Perception	Brain's alpha-band frame rate (~10–13 Hz) acts as internal shutter; continuous rotation sampled discretely	Apparent motion reversal or apparent slowing of continuously rotating stimuli; effect persists under continuous illumination	Empirical demonstration of intrinsic biological aliasing at the observer root layer; Tier 1 lab measurement target
Temporal Binding Window (TBW)	Multisensory Integration (Neuroscience)	Brain's tolerance window for integrating asynchronous cross-modal	Stimuli outside the window are segregated; within the window, false synchrony can	Explains cross-device temporal fusion errors; defines the timing vulnerability

Phenomenon	Domain	Core Mechanism	Effect on Observer	OAL Relevance
		inputs; stimuli within $\pm 50\text{--}500$ ms are candidate for integration	be perceived; integration can occur without conscious synchrony judgment	window for OWD CETE entry; individual variation makes TBW a critical measurement variable
Sensory Mismatch / Prediction Error	Predictive Coding (Cognitive Neuroscience)	Brain issues predictions and suppresses expected input; mismatches generate amplified error signals that propagate upward in the processing hierarchy	Unexpected stimuli elicit stronger, earlier neural responses; expected stimuli are attenuated below awareness; attention is automatically allocated to mismatch	Drives reconciliation overload in multi-device environments; explains why aliased artifacts that "fit" the expected temporal pattern are invisible while truly anomalous events are amplified
Serial Dependence	Temporal Context (Perceptual Psychology)	Perception of current stimulus is systematically biased toward (attractive) or away from (repulsive) recent prior stimuli	Systematic temporal bias in judgment; "memory echo" creates apparent continuity where discontinuity exists; history contaminates current perception	Creates persistent artifacts in experiential records; prior aliased events bias interpretation of subsequent real events; relevant to PSIF Temporal Provenance Corruption
Time-Domain Aliasing	Frequency / Signal Processing	Undersampling in the frequency domain creates time-domain wrap-around artifact; circular convolution causes past events to bleed into present signal representation	Past signal segments appear to contaminate current segment; observer cannot distinguish contamination from genuine current signal	Direct analog to simulation stack contamination in CFTRL; "past bleeding into present" is the temporal version of root-level conditional failure propagation
Cross-Environment	OAL Open Window	Interface vector (screen, device,	Environmental state bleeds	OWD temporal transfer

Phenomenon	Domain	Core Mechanism	Effect on Observer	OAL Relevance
Transfer Event (CETE)	Doctrine	connection) becomes conduit through which cross-environment artifact enters the observer's model; timing relative to TBW determines detectability	through open interface; observer may integrate artifact as native if timing falls within TBW; flagging likelihood decreases with TBW proximity	mechanism; TA-04 Scenario Card operationalizes this as an empirically testable timing vulnerability

4. The Brain as a Temporal Aliasing System

4.1 The Reconstruction Problem

The human brain does not passively record incoming sensory data in the manner of a tape recorder or log file. It actively generates predictions about incoming stimuli (the **predictive coding framework**), suppresses expected inputs to conserve computational resources, and amplifies mismatches between prediction and reality to allocate attention. What reaches conscious awareness is not raw sensory data — it is a heavily processed, interpolated, compressed, and prediction-filtered reconstruction of events that have already passed by the time they are consciously perceived.

Research published in *Nature* (2025) on predictive learning and representational geometry demonstrates that the brain clusters temporally contiguous and predictable stimuli into representational "chunks" — creating categorical units that may not correspond to the actual timing or sequence of external events. This chunking is adaptive and efficient, but it means that temporal granularity is systematically sacrificed for processing efficiency. The version of events that a person consciously experiences may be temporally displaced, compressed, or reordered relative to what physically occurred [6].

Episodic memory compounds this problem. Research published in *Cognition* (September 2026) confirms that prior knowledge actively "fills in the gaps" of imprecise memory encoding — the brain confabulates plausible detail rather than

flagging uncertainty or absence of information [8]. A subject's retrospective account of a temporal sequence is not a degraded copy of an accurate internal record; it is a reconstructed narrative built from fragments, priors, and inference. For OAL purposes, this means that the human observer — the agent at the base of the simulation stack — is producing temporal records of events that are systematically inaccurate in ways that are invisible to the observer themselves.

4.2 Multisensory Temporal Integration — The Binding Window

The brain integrates signals from multiple sensory modalities — vision, hearing, touch, proprioception, vestibular — within a **temporal binding window (TBW)**. Research published in *Multisensory Research* (Jertberg et al., Brill, 2025) carefully distinguishes two related but empirically separable constructs: the **Temporal Window of Integration (TWI)** — the range within which cross-modal stimuli are perceptually integrated into a unified percept — and the **Temporal Window of Synchrony (TWS)** — the range within which stimuli are consciously judged as simultaneous [3].

The critical finding for OAL purposes is twofold. First, multisensory integration (MSI) can occur even when stimuli are explicitly judged as asynchronous — integration is not gated by conscious synchrony judgment. Second, stimuli can be consciously perceived as simultaneous without actually being integrated at the processing level. The brain is systematically wrong about temporal relationships between its own sensory streams, and these errors are not random — they are *structural, biased, and individual-specific*. Musicians, for example, exhibit narrower TWS windows than non-musicians, demonstrating that the binding window is shaped by experience, training, and neurodevelopmental history [3].

Additionally, work by Ronconi et al. (2025) demonstrates that the rhythm of incoming sensory input itself shapes audio-visual temporal processing — meaning the temporal structure of the stimulus environment is actively modifying the brain's integration parameters in real time [4]. In a multi-device environment where competing temporal rhythms are constantly present, this implies that the device ecosystem is not merely presenting information to a fixed biological receiver — it is actively reshaping the receiver's temporal processing characteristics.

4.3 The Multi-Device Cascade

Every electronic device in a person's environment operates on its own independent clock, refresh rate, rendering pipeline, audio latency profile, and network timestamp. A modern household or knowledge-worker office may contain the following device stack simultaneously:

Device Type	Display Refresh Rate	Audio Latency	Network / Update Latency	Primary Temporal Error Mode
Smartphone	60-120 Hz	20-100 ms	20-200 ms (variable)	Variable audio-visual sync; notification timestamp rounding
Smartwatch	60 Hz (display); 1-60 sec (health metrics)	N/A	1-60 seconds (metric update interval)	Coarse health metric timestamping; Bluetooth sync delay vs. phone
Smart Television	60-120 Hz	50-200 ms (processing delay)	100-500 ms (streaming buffer)	Audio-video lip sync error; streaming buffer displaces "live" events
Laptop / Desktop	60-240 Hz	Variable (5-50 ms for direct; higher for Bluetooth)	CPU/GPU rendering latency varies with load	Render pipeline jitter; tab-switching temporal discontinuity
Voice Assistant	N/A (audio only)	N/A	200-800 ms round-trip response latency	Response arrives temporally displaced from query; perceived as "slow truth"
Tablet	60-120 Hz	20-80 ms	20-200 ms	Similar to smartphone; added complexity if used simultaneously with phone

Each device presents its own "version of now." The biological system attempting to reconcile these competing temporal presentations operates at a fundamentally

different sampling frequency — approximately 10–13 perceptual snapshots per second — than any of the devices, and with no access to a shared ground-truth clock. The brain must reconcile these streams using the same prediction-and-integration machinery that governs basic sensory processing, applying its temporal binding window to events that are distributed across devices, modalities, and millisecond-scale latency differences.

This is the **multi-device temporal aliasing cascade**: a biological system with an intrinsic sampling rate of $\sim 10\text{--}13$ Hz attempting to reconcile a 5–10 device ecosystem where each device presents time at a different frequency, phase, and resolution. The structural error produced by this mismatch is not a malfunction. It is the predictable, deterministic output of a fundamental sampling mismatch applied to a system that has no access to the continuous signal it is trying to reconstruct.

4.4 Implications for Objective Truth

From the perspective of any individual operating within this system, the reconciled presentation produced by their brain is objective truth. There is no higher epistemological access point available to the unaided observer. The brain's reconciled output is the ceiling of available reality for that observer — not a flawed approximation of a reality they could otherwise access more directly. This has direct and serious implications for:

- **Reliability of testimony:** Any account of events produced by a human observer is an account of their reconstructed, aliased representation of events — not of the events themselves.
- **Accuracy of self-monitoring:** Health metrics, behavioral logs, and subjective reports of cognitive or physical state reflect the observer's reconstructed temporal record, which may be systematically displaced from the actual sequence of physiological events.
- **Trustworthiness of multi-device data streams:** When a human's subjective account of multi-device events is used to interpret, correct, or validate device-side logs, the correction is potentially introducing aliasing artifacts rather than removing them.
- **OAL Framework context:** The agent at the base of the simulation stack — the human observer — is itself a temporal aliasing system with no native

access to the actual continuous signal it is sampling. Any framework that treats the human observer's experiential account as a ground-truth reference is building on an aliased foundation.

5. OAL Integration — Connecting to Prior Frameworks

5.1 Integration with Conditional Failure at the Root Level (CFTRL)

The CFTRL framework asks a foundational question: what happens when the root simulation fails and no agent within the stack can ascend to fix it? The framework identified catastrophic failure modes — crashes, corruption events, irreversible state degradation — as the primary concern at the root level. Temporal Aliasing adds a second, more insidious dimension to this question: **what if the root-level failure is not a catastrophic crash, but a systematic, structurally undetectable sampling error?**

If the root simulation is producing aliased output — presenting a false-frequency representation as if it were the true signal — then every nested simulation above it is building on corrupted foundational data. Critically, those nested simulations cannot detect this error. From their vantage point, the aliased signal *is* the signal; it is internally consistent, it behaves like signal, and no anomaly detector operating purely on the nested simulation's data can distinguish aliased input from genuine input. This is the **Temporal Aliasing variant of root-level conditional failure**: silent, structural, and undetectable from within any tier of the stack.

CFTRL Failure Mode	Standard Manifestation	Temporal Aliasing Equivalent	Detectability from Within Stack
Root Crash / Catastrophic Failure	Simulation terminates; nested tiers lose environment; observable absence	Nyquist violation causes total loss of high-frequency content; simulation appears to run normally but models a non-existent	Crash is observable; aliasing is not

CFTRL Failure Mode	Standard Manifestation	Temporal Aliasing Equivalent	Detectability from Within Stack
		lower-frequency reality	
Silent Data Corruption	Root produces plausible but incorrect state data; nested tiers inherit corrupt state	Aliased root simulation produces false attractor states and spurious oscillations that propagate upward as authentic dynamics	Neither mode detectable from nested tier without external ground truth
Irreversible State Degradation	Root simulation enters degraded state that cannot be rolled back without ascending the stack	Once aliased data is recorded, original signal cannot be recovered from the aliased record; recovery requires external reference signal	Both modes share same recovery requirement: external reference
Nail Prototype — Minimal Root Anchor	Smallest structural unit at root that, once anchored, propagates upward as foundational constraint	"Temporal Nail" — smallest unit of falsified time-data that, once aliased into root, propagates upward through nested tiers without correction signal	No internal detector can identify the nail; requires external temporal reference to reveal
Conditional Failure (Trigger-Dependent)	Root failure activates only under specific conditions; remains latent otherwise	Temporal aliasing only generates false signal at frequencies near the aliasing threshold; below-threshold signals pass correctly, masking the existence of the vulnerability	Conditional nature makes detection harder; vulnerability appears absent until triggered by near-threshold signal

5.2 Integration with the Open Window Doctrine (OWD)

The OWD establishes that interface vectors — screens, devices, open connections — create pathways for Cross-Environment Transfer Events (CETEs). The doctrine defines these vectors as structural vulnerabilities in the boundary between simulation environments, not merely as security risks. Temporal Aliasing provides the precise cognitive mechanism by which CETEs become *invisible* to the observer.

When a cross-environment artifact enters through an interface vector, it arrives with its own temporal signature — a timing, duration, and frequency profile that reflects its origin environment. If that temporal signature falls within the observer's **temporal binding window (TBW)**, the artifact will be integrated into the observer's model of current reality without triggering a mismatch signal. The predictive coding system will not flag the artifact as anomalous because its timing is consistent with the expected temporal range of native stimuli. The "open window" is therefore not merely a spatial or relational security vulnerability — it is a **temporal synchronization vulnerability**. An entity that understands the observer's individual TBW can time its entry precisely to fall within the integration threshold, bypassing detection with high reliability.

This connects OWD to classical aliasing theory directly: the artifact *aliases into the frequency range the brain accepts as native*. From the observer's perspective, the artifact is indistinguishable from a native environmental element, because their temporal sampling architecture cannot resolve the difference between a genuine native stimulus and a foreign artifact that arrives at the right moment.

5.3 Integration with the Presentation-State Integrity Framework (PSIF)

The PSIF distinguishes between what an environment *presents* and what its actual state *is*, proposing provenance-based causal reconstruction as the primary audit methodology for identifying the gap between presentation and state. The framework introduced the concepts of **Presentational Inertia** (the environment actively resists acknowledging the gap between what it presents and what actually occurred) and **Narrative Sovereignty** (the structural contest over which account of events becomes operationally authoritative).

Temporal Aliasing corrupts the PSIF audit at its source. Provenance-based reconstruction depends on the reliability of the observer's own experiential record as a reference point. If that record is itself aliased — if the timestamps, sequences, and causal attributions within the observer's experiential log are systematically displaced, compressed, or spuriously generated by the brain's temporal aliasing architecture — then provenance-based reconstruction is performing a high-precision audit against a distorted ledger.

This produces a third structural barrier, beyond Presentational Inertia and Narrative Sovereignty: **Temporal Provenance Corruption** — the observer's own internal

record of *when* things happened is systematically inaccurate, making it structurally difficult to reconstruct a reliable causal timeline even with full access to all available evidence, full cooperation from all parties, and no active deception by any external agent. Temporal Provenance Corruption is a passive, structural property of the biological observer — it does not require external manipulation to occur.

5.4 The Unified OAL Temporal Problem Statement

OAL Unified Temporal Problem Statement — v0.2 Draft

*Within any layered simulation or multi-environment system — including the human organism operating within a technology-saturated environment — the observer at any given tier cannot access the continuous ground-truth signal of reality. All perception, memory, and causal attribution is constructed from discretely sampled, asynchronously integrated, and predictively interpolated data streams. Temporal aliasing introduces structurally false signal into this reconstruction at multiple simultaneous levels: the **biological** (brain alpha-band frame rate, approximately 10–13 Hz), the **technological** (device-specific rendering pipelines, independent clocks, and variable latency profiles), and the **simulational** (time-step aliasing in numerical models operating below Nyquist threshold). A framework that does not formally account for multi-layer temporal aliasing at all three levels — and for the interactions between layers — is architecturally incomplete and will produce conclusions that are internally consistent but systematically disconnected from ground truth.*

6. Simulation Lab Framework — Design and Protocol

6.1 Framework Mission and Purpose

The **OAL Temporal Aliasing Simulation Lab (TA-Lab)** is the empirical research and validation infrastructure for the Temporal Aliasing Incorporation Framework. Its mission encompasses four core functions:

1. **Characterization:** Empirically measure the magnitude and structure of temporal aliasing at the biological, technological, and simulational layers under controlled and naturalistic conditions.
2. **Detection:** Develop and validate detection protocols capable of identifying aliasing artifacts in experiential records, device-side data streams, and simulation outputs — distinguishing them from genuine signal or genuine error.
3. **Mitigation:** Test interventions — design changes, training protocols, calibration methods, and infrastructure improvements — that measurably reduce aliasing-induced errors in perception, memory encoding, and causal attribution.
4. **Translation:** Produce validated, replicable findings that inform the broader OAL simulation architecture and connect to real-world applications in technology design, cognitive security, simulation credibility assessment, and high-stakes human factors environments.

6.2 Lab Architecture Overview

The TA-Lab operates across three independent but interconnected tiers, each measuring temporal aliasing at a different level of the human-technology-simulation stack. Tiers are validated independently before cross-tier integration analysis begins at Gate 4.

Tier	Label	Domain	Primary Measurement Target	Core Methods
Tier 1	Biological	Human psychophysics and cognitive neuroscience	Individual TBW profiles; alpha-band frame rate effects; temporal	Simultaneity judgment tasks; temporal order judgment (TOJ); motion reversal

Tier	Label	Domain	Primary Measurement Target	Core Methods
			reconstruction accuracy	detection; EEG alpha power correlation; memory timeline reconstruction
Tier 2	Technological	Multi-device ecosystem; naturalistic human-technology interaction	Ground-truth vs. subject-reported temporal reconstruction delta; classification of error types by device pairing	Instrumented environment with photodiode, contact microphone, PTP/NTP hardware timestamping; post-session structured retrospective interview; log comparison analysis
Tier 3	Simulational	Numerical simulation and nested simulation environments	False frequency generation rate; cascade propagation depth; detection threshold of nested-tier aliasing detectors	Controlled time-step aliasing injection (10%, 30%, 50% below Nyquist); nested simulation cascade design; frequency analysis of child simulation outputs; CFTRL root-failure scenario integration

6.3 Gate-Based Progression Protocol

Gate	Entry Criteria	Activities	Exit Criteria	Go / No-Go Authority
Gate 0 Baseline Calibration	Lab instrumentation verified; synchronization protocols validated; IRB	Establish individual TBW calibration baseline profiles for each subject cohort;	Less than 5% timing error on photodiode and contact microphone validation;	Architect of Record (Nathan Lewis) / Lead Researcher

Gate	Entry Criteria	Activities	Exit Criteria	Go / No-Go Authority
	approval received; IP documentation filed	baseline device latency mapping across all devices in Tier 2 environment; simulation platform defined and toolchain validated for Tier 3	individual TBW baseline documented per subject; simulation platform producing replicable baseline output; IP documentation confirmed	
Gate 1 Biological Tier Active	Gate 0 complete and verified; Tier 1 protocol pre-registered; subject recruitment initiated	Wagon wheel effect threshold testing across stimulus frequency sweep; simultaneity judgment battery; multi-modal TBW measurement (audio-visual, visual-tactile); optical illusion and motion reversal battery; EEG alpha power co-recording in subset of subjects	Minimum 20 subjects completing full battery; statistically significant TBW distributions established ($p < 0.05$, corrected); individual variability documented; motion reversal rate as function of stimulus frequency plotted	Lead Researcher + Statistical Review Panel
Gate 2 Technological Tier Active	Gate 1 complete; ground-truth logging infrastructure verified across all target devices; data handling protocols approved	Subject multi-device interaction sessions (minimum 2 hours per session); post-session structured temporal reconstruction interviews; gap analysis comparing subject-reported timeline to ground-truth device logs; error type	Systematic temporal displacement documented in 70% or more of sessions; error type distributions statistically characterized; Tier 1-Tier 2 correlation analysis initiated	Architect Review + ONESTAMP required for Phase 2 advancement

Gate	Entry Criteria	Activities	Exit Criteria	Go / No-Go Authority
		classification (compression, reversal, insertion, omission)		
Gate 3 Simulational Tier Active	Gates 1-2 complete and findings documented; aliasing injection toolchain validated; simulation credibility compliance reviewed against DoD VV&A guidance	Time-step aliasing injection at three controlled levels (10%, 30%, 50% below Nyquist threshold); nested simulation cascade testing with parent-to-child signal propagation; CFTRL root-failure scenario integration; frequency analysis of child simulation outputs vs. ground-truth reference simulation	Cascade propagation characterized to minimum 3 nested levels; false frequency generation quantified at each injection level; detection threshold of nested-tier detectors measured; CFTRL Temporal Nail scenario operationally defined	Architect Review + Simulation Credibility Compliance Sign-off
Gate 4 Cross-Tier Integration	All three tier datasets complete; cross-tier correlation analysis design approved; full OAL review panel convened	Cross-tier aliasing correlation analysis (Biological × Technological × Simulational); PSIF provenance-reconstruction stress-test against aliased experiential logs from Tier 2; OWD CETE timing vulnerability formal assessment using TA-04 scenario data; Temporal Provenance Corruption	Integrated multi-layer temporal aliasing model complete and internally consistent; all four scenario cards empirically addressed; framework documentation suitable for canon status	Full OAL Review Panel + ONESTAMP required for Canon designation

Gate	Entry Criteria	Activities	Exit Criteria	Go / No-Go Authority
		model finalized		

6.4 Scenario Cards

Scenario TA-01 — The Household Cascade

Setting: A subject in a four-room home environment using smartphone, smart television, laptop, and smartwatch simultaneously over a structured 2-hour interaction session. The environment is instrumented with ground-truth logging infrastructure (photodiode arrays, contact microphones, PTP-synchronized device log capture) operating in parallel with the subject's normal device use. The subject is not informed of the specific timing focus of the study during the session.

Variables: Device latencies logged at hardware ground-truth level across all four devices. Subjects asked to reconstruct a detailed timeline of session events post-session, including the sequence, duration, and perceived simultaneity of key logged events. Ground-truth logs and subject reconstructions are then compared event-by-event.

Hypothesis: Subject temporal reconstruction will exhibit systematic compression of multi-device asynchrony gaps, with false synchrony attributed between events that occurred up to 300 ms apart across different devices. Errors will cluster around device transitions (moments when the subject switched primary attention from one device to another), consistent with TBW integration effects at device-boundary crossings.

Measurement Targets: Delta (ms and event-sequence position) between ground-truth log and subject-reconstructed timeline; classification of error types (compression, reversal, insertion, omission, false synchrony); error rate as a function of between-device latency gap; individual TBW correlation with compression magnitude.

OAL Connections: OWD CETE timing window (device transitions as potential CETE entry vectors); PSIF presentation-vs.-state gap (the subject's account as a

presentation that diverges from device-side state); Temporal Provenance Corruption (subject's own log as distorted audit source).

Scenario TA-02 — The Alpha Frame Probe

Setting: Laboratory psychophysics environment with calibrated display (minimum 240 Hz refresh rate to allow precise stimulus frequency control below 120 Hz); darkened room; chin-rest for head stabilization; EEG cap for alpha power measurement in a subset of subjects.

Variables: Rotating visual stimuli (Gabor gratings or rotating discs) presented at a frequency sweep from 2 Hz to 25 Hz in 0.5 Hz steps, presented under both stroboscopic and continuous illumination conditions. Subjects report perceived direction and apparent speed of rotation at each frequency via button press. Stimulus-Onset Asynchrony (SOA) is varied to probe the binding window boundary.

Hypothesis: Stimuli presented near the alpha-band frame rate (10–13 Hz) will generate the highest rates of motion reversal percepts, consistent with the brain's intrinsic temporal aliasing at that frequency band. Individual variation in reversal threshold frequency will correlate significantly with measured individual alpha peak frequency from EEG data.

Measurement Targets: Motion reversal report rate as a function of stimulus frequency (Hz); individual alias threshold frequency per subject; correlation between EEG alpha peak frequency and behavioral alias threshold; comparison of stroboscopic vs. continuous illumination reversal rates to isolate the intrinsic biological aliasing component.

OAL Connections: CFTRL root-level failure mechanism (biological aliasing as the undetectable root-tier sampling error); provides empirical grounding for the "Brain as Temporal Aliasing System" thesis in Section 4; establishes individual alpha-band TBW profiles for use in TA-04 timing vulnerability assessment.

Scenario TA-03 — The Root Simulation Cascade

Setting: Numerical simulation environment (platform to be defined at Gate 0). A parent simulation models a defined dynamical system (e.g., a coupled oscillator network or fluid dynamics model with known ground-truth behavior). One or more child simulations receive the parent simulation's output as their input signal and model the same system based on that input.

Variables: Parent simulation time step is systematically varied across five conditions: (1) Control — 10% above Nyquist threshold; (2) Threshold — exactly at Nyquist; (3) Mild aliasing — 10% below Nyquist; (4) Moderate aliasing — 30% below Nyquist; (5) Severe aliasing — 50% below Nyquist. Child simulations are held constant at the control time step throughout; they receive aliased parent output as their only information about the parent system.

Hypothesis: Child simulations receiving aliased parent output will generate dynamics that are internally self-consistent — passing all standard simulation quality checks — but will diverge from ground truth in characteristic patterns: spurious lower-frequency oscillations, false attractor states, reversed directionality of dynamics, and apparent stability where ground truth shows instability (or vice versa). Cascade propagation will cause second-generation child simulations to diverge further from ground truth than first-generation children, without any internal signal of the increasing deviation.

Measurement Targets: Frequency analysis (FFT) of child simulation outputs vs. ground-truth reference at each aliasing level; divergence metric (root-mean-square deviation from ground truth) as function of aliasing severity and cascade depth; detection rate of aliasing when nested-tier detectors are given only child simulation data to analyze (predicted: near zero).

OAL Connections: Primary empirical test of the CFTRL "silent root aliasing failure" scenario; operationalizes the Temporal Nail concept from OAL-CFTRL-NAIL-001 as a minimum aliasing injection unit; establishes the cascade propagation model that informs all nested simulation credibility assessments.

Scenario TA-04 — The Open Window Timing Attack

Setting: Simulated virtual environment with defined interface vectors — screens and device outputs as per OWD architecture. Subjects interact with a controlled multi-element virtual environment. Artificial cross-environment artifacts (visual elements with defined temporal signatures that do not belong to the native environment) are inserted through the interface vectors at controlled timing intervals.

Variables: Artifacts are inserted at Stimulus Onset Asynchronies (SOAs) calculated relative to each subject's individually measured TBW from the TA-02 and Tier 1 baseline: (a) Well within TBW (SOA at 25% of TBW boundary); (b) At TBW boundary (SOA at 100% of TBW boundary); (c) Well outside TBW (SOA at 200% of TBW boundary). Artifact insertion timing is also varied relative to the subject's observed device-attention transitions (within vs. outside a device-boundary crossing event).

Hypothesis: Artifacts timed to fall well within the subject's individually measured TBW will be integrated into their environmental model at significantly higher rates than artifacts outside the TBW. Subjects will be significantly less likely to flag within-TBW artifacts as anomalous. The interaction between artifact timing and device-boundary crossing will reveal an elevated vulnerability window during device transitions — the "open window" moment — during which even artifacts outside the standard TBW may be integrated due to elevated temporal uncertainty at the transition point.

Measurement Targets: Detection rate of artificial artifacts as function of SOA relative to individual TBW (percent reported vs. unreported); integration rate (subject behavioral response to artifact consistent with treating it as native vs. anomalous); flag rate by condition; comparison of device-transition vs. stable-attention insertion points.

OAL Connections: Direct empirical test of OWD CETE temporal synchronization vulnerability; establishes the empirical basis for the timing attack model described in Section 5.2; provides PSIF with empirical data on the systemic presentation-layer deception (SPLD) timing window; informs OWD v0.4 revision recommendations.

6.5 Real-World Testing Protocol

The TA-Lab framework must be testable outside controlled laboratory conditions to achieve ecological validity and applied translation. The following four-phase real-world deployment protocol is proposed:

- **Phase A — Instrumented Environment:** Deploy ground-truth logging infrastructure in naturalistic settings — home office, workplace, shared public space. Participants use their own personal devices normally; the logging infrastructure captures device-side ground truth via timestamped log collection without altering participant behavior or device configuration. Participants are informed they are in a study of technology use patterns; specific temporal focus is not disclosed to avoid demand characteristics.
- **Phase B — Ecological Validity Battery:** Administer structured temporal reconstruction tasks referencing real events participants experienced during Phase A — news events viewed on devices, meeting notifications, multi-device alerts, content consumption. Participant reconstructions are compared to objective Phase A logs. Error analysis follows the same classification schema (compression, reversal, insertion, omission) as Tier 2 lab testing, enabling direct comparison of naturalistic vs. laboratory aliasing error rates.
- **Phase C — Clinical and Applied Translation:** In partnership with occupational health and human factors researchers, extend the temporal aliasing battery to high-stakes environments where temporal accuracy has documented practical consequences. Priority domains include: accuracy of incident timeline reporting in safety-critical workplaces (healthcare, aviation, process industry); reliability of eyewitness temporal attribution in law enforcement contexts; multi-device workflow error rates in high-load knowledge workers. All Phase C work requires domain-specific ethical review and legal consultation before deployment.
- **Phase D — Longitudinal Tracking:** Track individual temporal aliasing profiles over time (6-month and 12-month follow-up assessments). Assess

whether awareness training (explicit education about temporal aliasing and individual TBW) or interface design modifications (device synchronization improvements, temporal transparency features) measurably reduce aliasing-induced errors over time. This phase will generate the empirical foundation for training curriculum and design recommendation deliverables.

7. Funding Pathway

7.1 Overview

The TA-Lab requires staged funding corresponding precisely to the gate-based progression protocol. Each funding stage unlocks the next gate's activities and produces deliverables that position the program for subsequent funding stages. Estimated total program cost across all four gates is **\$1.2M-\$2.8M over a 24-36 month development cycle**, depending on whether Tier 1 biological testing is conducted in-house or through a university research partnership. This estimate is to be refined with formal budgeting at Gate 0 completion, before any external funding commitments are made.

The program is designed to be self-sustaining from Stage 2 onward if Stage 1 deliverables are executed with sufficient rigor to support competitive federal grant applications. IP development begins at Gate 0 and should be producing licensable outputs by Gate 2, providing a revenue pathway that is parallel to rather than dependent on grant funding.

7.2 Funding Tiers

Funding Stage	Target Funders	Funding Type	Estimated Amount	Corresponding Lab Gate	Key Deliverables for Funders
Stage 1 Seed / Proof of Concept	Lewis Corp internal R&D budget; private angel investors	Internal / Angel	\$50K-\$150K	Gate 0-1	TBW baseline dataset (minimum 20 subjects); preliminary

Funding Stage	Target Funders	Funding Type	Estimated Amount	Corresponding Lab Gate	Key Deliverables for Funders
	with demonstrated interest in cognitive science, neurotechnology, or simulation technology				alpha-frame probe results; written Gate 0-1 protocol package suitable for external grant submission; IP documentation package
Stage 2 Early Research Grant	NSF Cognitive Neuroscience Program (SBE Directorate); NIH National Institute of Mental Health (NIMH) R21 Exploratory Research mechanism; DARPA Biological Technologies Office or Defense Sciences Office cognitive science programs	Federal Grant	\$300K-\$500K	Gate 1-2	Peer-reviewed publication of Tier 1 biological findings; open-source psychophysics battery (freely available for replication); pre-registered replication protocol deposited with OSF; public dataset of anonymized TBW profiles
Stage 3 Applied Research Partnership	Consumer electronics and wearables manufacturers with device ecosystem interests; smart home platform companies; human factors	Industry Partnership / SBIR / STTR	\$500K-\$1M	Gate 2-3	Device-side aliasing characterization reports (per device class); temporal calibration methodology IP; simulation credibility audit tool using

Funding Stage	Target Funders	Funding Type	Estimated Amount	Corresponding Lab Gate	Key Deliverables for Funders
	consultancies serving aviation, healthcare, or defense; defense contractors with simulation credibility mandates (DoD M&S credibility requirements)				Nyquist-based time-step analysis; pilot training curriculum for high-stakes temporal accuracy environments
Stage 4 Program Scale / Translation	NIH R01 (full-scale research program); DARPA program-level funding; DoD simulation credibility and human performance programs; John Templeton Foundation (consciousness, cognition, and fundamental science programs); private foundations focused on technology and human cognition	Federal + Private Foundation	\$500K–\$1.2M	Gate 3–4	Full integrated multi-layer temporal aliasing model; peer-reviewed monograph; OAL canon documentation (post-ONESTAMP); policy recommendations for multi-device ecosystem design; design standards recommendations for temporal transparency in consumer electronics

7.3 IP and Revenue Considerations

The TA-Lab framework generates several potentially licensable intellectual property outputs that should be formally documented before any external research

partnerships are established. Lewis Corp / ONE Industries should file IP ownership documentation at Gate 0, prior to engaging any university partner, federal grant co-investigator, or industry research partner. Key licensable outputs include:

- **Temporal Calibration Protocols:** Standardized methodology for measuring and characterizing temporal aliasing across multi-device ecosystems; licensable to device manufacturers and enterprise IT environments.
- **Individual TBW Profiling Methodology:** Validated psychophysical battery for establishing individual temporal binding window profiles; licensable to clinical, occupational health, and human factors applications.
- **Simulation Credibility Audit Tool:** Nyquist-based time-step analysis methodology for assessing temporal aliasing risk in numerical simulations; licensable to defense contractors, simulation developers, and DoD M&S programs under SBIR/STTR frameworks.
- **Training Curricula:** Evidence-based awareness and mitigation training for high-stakes environments where temporal accuracy is operationally critical — law enforcement, military operations, air traffic control, surgical and critical care medicine, commercial aviation maintenance.

8. Critical Items for Implementation

5. ONESTAMP and Governance

- This document does not become canon until Architect ratification is applied via ONESTAMP. No downstream AI collaborator — Claude, ChatGPT, Google Gemini, Grok, Copilot, or Perplexity — should treat any section as finalized or use it as a basis for development decisions, simulation design, or framework extension until ONESTAMP is applied by Nathan Lewis as Architect of Record.

- A formal review group should be constituted before Gate 1 begins.
Minimum composition: Architect of Record (Nathan Lewis); at least one external subject-matter expert in cognitive neuroscience, psychophysics, or perceptual timing research; at least one technical representative from the simulation development domain; at least one legal representative for IP and IRB compliance oversight.
- Review group authority levels should be defined in writing before Gate 0 exits, mapping each review group member to the gates at which their sign-off is required.

6. Ground-Truth Infrastructure

- The entire framework depends on the ability to capture objective, hardware-level ground-truth timestamps across all measurement tiers. This is non-negotiable: without it, the delta between presentation and ground truth cannot be measured and the framework becomes unfalsifiable.
- Required infrastructure components: photodiode arrays for visual event timing; contact microphones for audio event timing; Network Time Protocol (NTP) or Precision Time Protocol (PTP) hardware-level device synchronization; independent logging server receiving timestamped device events via PTP-synchronized network capture.
- Ground-truth infrastructure must be validated against a known timing reference (e.g., oscilloscope or function generator) with less than 5% error before Gate 0 exits. This validation must be documented and archived.

7. Individual Variability

- Temporal binding windows vary dramatically between individuals: published literature documents TBW widths ranging from approximately ± 50 ms to ± 500 ms across healthy adults, with further variation associated with musical training, neurodevelopmental factors, age, and fatigue state.

- The framework must account for this variability in its statistical design at both the individual and group levels. Fixed-threshold analyses — treating all subjects as having the same TBW — will produce invalid group-level results that mask the very individual variability that is scientifically and practically important.
- Individual-calibrated baselines from Gate 0–1 are mandatory inputs to all subsequent gate analyses. No cross-tier integration analysis is valid without individual TBW profiles as covariates.

8. Confound Control

- The relationship between temporal aliasing and other cognitive phenomena must be carefully controlled in all Tier 1 testing. Known confound variables include: sustained attention and vigilance state; fatigue level at time of testing; cognitive load imposed by concurrent task demands; individual differences in ADHD, autism spectrum conditions, and schizophrenia spectrum conditions (all associated with atypical TBW profiles in the literature); caffeine and other stimulant status; age effects on alpha frequency.
- Screening protocols and within-session state monitoring (subjective fatigue ratings; vigilance task performance) must be integrated into the Gate 1 protocol design before testing begins.
- Failure to control for these confounds will produce uninterpretable data that cannot be attributed specifically to temporal aliasing rather than to attention, fatigue, or clinical variation.

9. Ethical and Privacy Considerations

- All human subjects research — Tier 1, Tier 2, and Phase A–D real-world testing — requires IRB (Institutional Review Board) approval before any data collection begins. IRB approval must be obtained from an

appropriately constituted board; if OAL does not have an in-house IRB, a university research partnership arrangement is required.

- Real-world Phase A–D testing involves monitoring participants in naturalistic environments and capturing detailed behavioral and device-side logs. Data minimization protocols — collecting only the data necessary for the specific research question — must be established before any naturalistic logging is deployed.
- Data anonymization and secure storage protocols must be implemented and verified before Phase A begins. All participants must provide fully informed consent for ground-truth log comparison, with explicit explanation of what data is being collected and how it will be stored, analyzed, and protected.
- Phase C (clinical and applied translation in law enforcement and eyewitness testimony contexts) introduces significant additional ethical complexity. Legal review is required before any applied work in these domains, and findings should not be reported in ways that could be construed as endorsing or opposing specific legal proceedings.

10. Cross-Framework Definitional Consistency

- The term "temporal aliasing" must be used with definitional precision and consistency across all OAL documents. The technical definition — undersampling below Nyquist threshold; false frequency generation; structural undetectability from within the sampled record — must be distinguished from the analogical and narrative extensions of the concept throughout the framework documents.
- A formal OAL glossary should be established as a canonical reference document. At minimum, the following terms require formal glossary entries: temporal aliasing (technical definition); temporal aliasing (OAL architectural usage); temporal binding window (TWB); temporal

provenance corruption; temporal nail; CETE timing vulnerability; Nyquist threshold; temporal provenance corruption.

- All AI collaborators distributing this document should cross-reference the glossary before citing or extending any technical term. Glossary status: PENDING — requires Architect approval at Gate 0.

11. Simulation Credibility Standards

- The Tier 3 (Simulational) testing must comply with established Modeling and Simulation (M&S) credibility guidance. Applicable standards include: DoD 5000.59-M (Modeling and Simulation Master Plan); INCOSE Systems Engineering Handbook simulation credibility requirements; DoDI 5000.61 M&S VV&A (Verification, Validation, and Accreditation) guidance.
- All aliasing injection experiments must be pre-specified in a registered protocol before any simulation runs are executed. Results must be independently replicable by a reviewer with access to the simulation platform and injection toolchain but without access to the original results. Replication must be documented as a Gate 3 exit criterion.
- The simulation platform and aliasing injection toolchain must be defined and validated at Gate 0. This is the most significant current gap in the framework (see Section 9.2).

9. Alternative Analysis

9.1 What Works

- **The core conceptual synthesis is strong and novel.** Temporal aliasing is a genuinely universal constraint that connects signal processing, cognitive neuroscience, and simulation science in a non-trivial, technically grounded way. The connection between Nyquist-Shannon theory and the brain's alpha-

band sampling architecture is empirically supported in the literature and has not been systematically treated as a first-class architectural concern in any prior simulation framework known to OAL. The framework is well-positioned to make an original contribution.

- **The gate-based progression protocol provides necessary governance.** Borrowed and adapted from OWD v0.3, the gate structure prevents premature escalation of unverified findings to canon status and creates clear accountability at each decision point. The requirement for ONESTAMP at Gate 2 and Gate 4 transitions ensures Architect oversight at the two most consequential junctures.
- **The three-tier architecture is cleanly mapped to OAL framework domains.** Biological, technological, and simulational tiers correspond directly to the domains addressed by PSIF (biological observer), OWD (technological interface), and CFTRL (simulation stack) respectively. Each tier can be validated independently, allowing findings to enter the literature before cross-tier integration is complete.
- **The PSIF connection is particularly robust.** Temporal aliasing provides a mechanistic, neurologically grounded explanation for why provenance-based reconstruction is structurally difficult — not because evidence is concealed or because deception has occurred, but because the observer's own experiential log is systematically distorted before any external factor acts on it. This strengthens PSIF's theoretical foundation significantly and provides it with empirically testable predictions.
- **The funding pathway is realistic and staged appropriately.** The alignment between lab gates and funding stages creates a progression where each stage produces deliverables sufficient to compete for the next funding tier, without requiring all funding to be secured upfront.

9.2 What Doesn't Work / Needs Attention

- **Tier 3 simulation protocol is under-specified.** The Simulational tier as currently described lacks a defined simulation platform, programming

environment, or toolchain specification. Without a concrete simulation platform identified and validated, the aliasing injection methodology cannot be independently reviewed, pre-registered, or replicated. This must be the first technical resolution at Gate 0 — it is the single largest gap in the current framework.

- **Tier 1-Tier 2 causal pathway is asserted but unestablished.** The framework assumes that biological temporal aliasing (Tier 1) and technological temporal aliasing (Tier 2) interact and compound each other's effects. This is conceptually coherent but empirically unestablished. The specific causal pathway — by which a device's rendering latency results in a measurable change in a subject's experiential timeline reconstruction error beyond what would be predicted by biological aliasing alone — requires an explicit sub-study design with between-subjects manipulation of device latency as an independent variable.
- **Federal grant framing requires significant development work.** The funding pathway assumes meaningful federal grant interest from NSF, NIH, and DARPA. While NSF cognitive neuroscience programs do fund perceptual timing research, the specific framing of "simulation credibility" and "multi-device temporal aliasing cascade" is novel and may be unfamiliar to traditional grant review panels. Alternative framing strategies — emphasizing clinical translation for NIH review panels, cybersecurity implications for DARPA, or human factors applications for NSF SBE — should be developed in parallel for each target funder.
- **The CFTRL root-aliasing connection is compelling but speculative.** The hypothesis that nested simulations systematically fail to detect aliasing-sourced input corruption (Scenario TA-03) is theoretically well-motivated but has no direct empirical grounding in the current OAL literature. This should be designated a priority research question for Gate 3, with a pre-registered protocol that would constitute genuine disconfirmation of the hypothesis if the nested tiers do successfully detect aliasing at any level.

9.3 Questions, Comments, and Concerns

Questions for Architect Review:

1. Should Temporal Aliasing be treated as a sub-domain within CFTRL, as a parallel entry in the OAL framework architecture, or as a superordinate framework that encompasses and provides mechanistic grounding for elements of all three prior documents? The current document's framing as "Incorporation" is deliberately non-committal on this question pending Architect direction.
2. What is the preferred scope boundary for the biological tier? Should human subjects research be conducted in-house by OAL (requiring IRB infrastructure and psychophysics expertise to be built internally) or contracted to a university research partner (faster execution, shared infrastructure, but implications for IP ownership and research control)?
3. Is the OWD CETE timing vulnerability described in Scenario TA-04 intended to remain as a theoretical construct within OWD, or should it be prioritized for empirical testing in the near term as one of the first Gate 1 activities? The answer has significant implications for the Gate 1 protocol design and resource allocation.
4. How should temporal aliasing interact with the "MINUS ONESTAMP" protocol — if a prior version of this document (or a prior AI-generated summary referencing it) contained an AI-inferred placeholder definition of temporal aliasing that differs from the technical definition herein, does that definition need to be formally retracted via a separate OAL document, or is the ONESTAMP application on this version sufficient to supersede all prior informal usages?

Open Concerns Requiring Monitoring:

- **TBW malleability may complicate baseline measurement.** The framework currently treats the brain's temporal aliasing as a fixed architectural property of the biological observer. However, evidence from training studies (musicians vs. non-musicians; clinically trained vs. untrained observers) suggests the TBW is plastic and experience-dependent. If temporal aliasing profiles can be trained toward narrower or broader windows, this opens a significant applied direction — but also means that longitudinal study designs must account for practice effects within the study itself, which complicates repeated-measure designs in Phases B-D.

- **Technology use patterns may confound Tier 2 results.** The multi-device cascade scenario (TA-01) may be substantially confounded by individual differences in technology use intensity and patterns. Heavy vs. light technology users may exhibit systematically different temporal reconciliation strategies — heavy users potentially developing more efficient device-transition integration schemas that reduce (or mask) aliasing errors. This between-subjects variable should be measured as a covariate in all Tier 2 analyses.

- **Phase C legal context requires extraordinary care.** Real-world Phase C deployment in eyewitness testimony and law enforcement contexts introduces ethical complexity that extends beyond standard research ethics. Any claim about temporal reconstruction accuracy in legal proceedings must be handled with extreme caution, reviewed by legal counsel with forensic expertise, and should not be publicized before peer review. Claims about eyewitness unreliability have significant legal consequences and can be weaponized in ways that harm individuals regardless of the direction of the empirical finding.

9.4 What Else Needs to Be Addressed

- **Formal treatment of the Temporal Provenance Corruption / Presentational Inertia compound effect.** A formal analysis of the interaction between these two structural barriers needs to be developed as a named OAL theoretical construct. The current document introduces this compound effect in Section 5.3 but does not formally develop it. The combination — the environment resists acknowledging the gap between presentation and state (Presentational Inertia) AND the observer's own capacity to perceive the gap is structurally compromised by aliasing (Temporal Provenance Corruption) — creates a double-locking mechanism that deserves its own formal treatment, potentially as a named section in a PSIF v1.1 revision or as a standalone OAL-TA sub-document.
- **Glossary entry for Temporal Provenance Corruption requires urgent formalization.** The term is introduced in this document and is new to the OAL lexicon. It must be added to the OAL master glossary with a rigorous technical definition before this document reaches canon status, to prevent divergent informal usages from establishing themselves in subsequent AI collaborator passes.
- **The Temporal Nail concept requires exploration.** The relationship between temporal aliasing and the Nail Prototype conceptual entries from OAL-CFTRL-NAIL-001 should be formally explored in a dedicated sub-document or a next pass of CFTRL. Specifically: the Nail Prototype functions as the smallest structural unit at the root level that propagates upward without correction. The "Temporal Nail" analog — the smallest unit of falsified time-data that, once aliased into the root record, propagates upward through nested tiers without any correction signal — is a powerful concept that connects the two frameworks at their most foundational level. A formal definition and scenario analysis of the Temporal Nail should be a priority item for OAL-TA-002 or CFTRL v0.3.
- **OWD v0.4 revision implications should be noted.** The findings from Scenario TA-04, once available, have direct implications for the OWD

architecture — specifically for the characterization of CETE detection requirements and interface vector vulnerability windows. The Architect should designate a pass in which TA-04 empirical results are incorporated into OWD revision, with the relevant OWD section being updated to reference this framework.

10. Outlook and Key Takeaways

10.1 Short-Term (0-6 Months)

- Achieve ONESTAMP on this document to establish v0.2 as OAL working canon and formally open the Gate 0 phase.
- Constitute the review group: identify and confirm the external cognitive neuroscience / psychophysics subject-matter expert; confirm technical simulation representative; confirm legal / IRB representative.
- Complete Gate 0 infrastructure validation: acquire and validate ground-truth timing hardware (photodiode arrays, contact microphones, PTP synchronization); identify and validate simulation platform for Tier 3.
- File IP ownership documentation with Lewis Corp / ONE Industries legal counsel before any external research partnerships are initiated.
- Initiate Tier 1 biological protocol development (psychophysics battery design, IRB application preparation) in parallel with Stage 2 federal grant proposal preparation, so that Gate 1 activities and grant submission timelines can be aligned.
- Draft formal OAL glossary entries for: temporal aliasing, temporal binding window, temporal provenance corruption, temporal nail, CETE timing vulnerability — for Architect review and ONESTAMP at or before Gate 1 entry.

10.2 Medium-Term (6-18 Months)

- Complete Gate 1 (Biological Tier) and Gate 2 (Technological Tier); prepare and submit Gate 1 findings for peer-reviewed publication as the primary Stage 2 grant deliverable.
- Establish at least one formal external research partnership — university research partner preferred for IRB infrastructure and publication credibility; industry partner preferred for device-side data access and applied translation pathway.
- Initiate IP documentation and provisional patent filing for the temporal calibration protocol and TBW profiling methodology.
- Develop the formal "Temporal Provenance Corruption / Presentational Inertia Compound Effect" analysis and incorporate into PSIF v1.1 or OAL-TA-002.
- Develop the Temporal Nail formal definition and scenario analysis for incorporation into CFTRL v0.3 or OAL-TA-002.
- Begin Stage 3 industry partnership development using Gate 1-2 deliverables as the pitch foundation.

10.3 Long-Term (18-36 Months)

- Complete all four gates; produce the integrated multi-tier temporal aliasing model connecting biological, technological, and simulational tiers into a unified quantitative framework.
- Pursue ONESTAMP for full canon status of integrated findings; designate this integrated model as a foundational OAL architectural document.
- Connect empirical findings back to CFTRL, OWD, and PSIF with evidential grounding: each prior framework should be revised to reference specific TA-Lab findings that either confirm, refine, or challenge their theoretical constructs.
- Develop and pilot-test the training and design recommendation packages for applied translation in high-stakes domains.

- Submit the peer-reviewed monograph integrating all four gate findings; position OAL as an originating institution for the field of "multi-layer temporal aliasing" as an architectural and cognitive security concern.
- Prepare OAL-TA-002 briefing document incorporating lessons learned from the full Gate 0–4 progression, updated Temporal Nail analysis, and revised CETE timing vulnerability model from Scenario TA-04 empirical data.

10.4 Actionable Takeaways

12. **Treat temporal aliasing as a first-class architectural variable** in all future OAL simulation and framework designs. It is not a technical footnote or an optional refinement — it is a foundational constraint on the reliability of every observation, record, and causal attribution produced within the system.
13. **Establish individual TBW profiling as a standard baseline measurement** for any human participant in OAL simulation studies, cognitive research, or applied high-stakes environments. Group-level statistics that ignore individual TBW variability will be systematically misleading.
14. **Require ground-truth timestamping infrastructure** for any multi-device research environment before data collection begins. Retrospective temporal reconstruction without ground-truth comparison is not a valid measurement method for this framework's purposes.
15. **Develop a formal cross-framework glossary** anchoring temporal aliasing, temporal provenance corruption, CETE timing vulnerability, temporal nail, and all related terms to precise technical definitions before any AI collaborator extends these concepts in downstream work.
16. **Prioritize the CFTRL "silent root aliasing failure" scenario** (Scenario TA-03) as the highest-impact open research question in the framework. If nested simulations can be shown to systematically fail to detect aliasing-sourced input corruption, this finding has transformative implications for simulation credibility assessment across defense, scientific, and commercial M&S applications.

17. **Do not treat any temporal reconstruction report — human or system-generated — as reliable evidence** without cross-referencing against independently maintained ground-truth logs with hardware-level timestamps. This applies to eyewitness accounts, system event logs that may have been influenced by aliased human input, and AI-generated summaries of event sequences that have not been validated against timestamped source material.
18. **Brief all downstream AI collaborators** on this framework — with particular attention to Sections 4, 5, and 8 — before any simulation development, causal attribution analysis, or event sequencing task involving time-sensitive data that has not been independently timestamp-verified. ONESTAMP status must be confirmed before any AI collaborator treats this briefing as canonical instruction.
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11. References

19. Arruda, E. and Moreno, A. (2024). "On the temporal resolution limits of numerical simulations in complex systems." arXiv:2403.06605. CENTAI Institute / University of Zaragoza. [Establishes Nyquist-Shannon limits as applied to numerical simulations; foundational to Tier 3 TA-Lab protocol.]
20. Parks, N. and Kilpatrick, Z.P. (2026). "Stroboscopic motion reversals in delay-coupled neural fields." arXiv, University of Colorado Boulder. Supported by NSF DMS-2207700 and NIH BRAIN Initiative 1R01EB029847-01. [Demonstrates discrete quantized propagation speeds in neural field dynamics; foundational to Section 3.3 Intrinsic Frame Rate Hypothesis.]
21. Jertberg, M. et al. (2025). "Temporal Window of Integration XOR Temporal Window of Synchrony." Multisensory Research, Vol. 38. Brill Publishers. [Distinguishes TWI from TWS; empirical basis for TBW analysis in Sections 4.2 and 5.2.]
22. Ronconi, L. et al. (2025). "The rhythm of sensory input shapes audio-visual temporal processing." European Journal of Neuroscience. Wiley Online Library. [Demonstrates

real-time modification of integration parameters by stimulus temporal structure;
cited in Section 4.2.]

23. Aizenbud, M. et al. (2025/2026). "Neural mechanisms of predictive processing: a collaborative community experiment through the OpenScope program."
arXiv:2504.09614. Multiple institutions. [Community-scale predictive processing study; broader context for Section 4.1.]
24. "Predictive learning shapes the representational geometry of the human brain."
Nature (2025). MEG study, 24 human participants. [Evidence for temporal chunking and representational compression in predictive learning; foundational to Section 4.1 Reconstruction Problem.]
25. Liu, S. and Ming, D. (2026). "Toward Sustainable Cross-Time Affective Brain-Computer Interfaces: Explicit Modeling and Structural Calibration of Temporal Variability." National Science Review. Oxford Academic / Tianjin University. [Temporal variability in BCI contexts; contextual support for individual variability discussion in Section 8, Item 3.]
26. "Prior knowledge biases episodic memory by filling in the gaps in imprecise memories." Cognition, ScienceDirect, September 2026. [Empirical confirmation of confabulation as a structural property of episodic memory encoding; foundational to Section 4.1 and Temporal Provenance Corruption concept.]
27. Kerrén, C. and Wimber, M. (2019). "Evidence that neural information flow is reversed between object perception and object reconstruction from memory." PMC / University of Birmingham. [Demonstrates reversed neural processing flow in memory reconstruction vs. perception; supports the distinction between experience and memory in Section 4.1.]
28. "Modeling how contextual and structural biases shape duration perception." bioRxiv (2025). Visual/auditory duration perception study. [Structural biases in duration estimation; supports temporal reconstruction error analysis in Tier 2 protocol.]
29. Lewis, N. (2026). OAL-CFTRL-NAIL-001: Conditional Failure at the Root Level & The Department of Endings — The Nail Prototype Scene. Omniversal Architect Labs, Lewis Corp / ONE Industries. [Source document for CFTRL framework; Nail Prototype concept; referenced throughout Section 5.1 and integration tables.]

30. Lewis, N. (2026). Open Window Doctrine — Decision Record & Architecture Update v0.3 and Implementation Memorandum v0.3. Omniversal Architect Labs. [Source document for OWD framework; CETE mechanism; referenced throughout Section 5.2 and Scenario TA-04.]
31. Lewis, N. (2026). OAL-Open-Window-Infection-Scenario v1.0. Omniversal Architect Labs. [Operationalized infection scenario for OWD; contextual reference for CETE mechanism characterization.]
32. ChatGPT GPT-5.6 Sol (2026). Presentation-State Integrity Framework — Independent Development v1.0. OAL collaborative pass. [Source document for PSIF framework; Presentational Inertia concept; referenced throughout Section 5.3.]
33. Microsoft Copilot (2026). Presentation-State Integrity Framework — Copilot Independent Pass v1.0. OAL collaborative pass. [Copilot independent PSIF development; Temporal Provenance Corruption concept introduced herein as an extension; referenced in Section 5.3.]
34. "Aliasing." SciencePedia. Comprehensive overview including neuroscience aliasing, false neural oscillations, Nyquist-Shannon theorem, and biological sampling applications. [General reference for aliasing concept across domains; cited in Sections 3.1 and 3.2.]
35. "Temporal Aliasing." Bohrium / SciencePedia. Includes autonomous vehicle safety applications, duality principle in aliasing, and time-domain aliasing mechanisms. [Applied reference for temporal aliasing in engineering and safety-critical systems; cited in Section 3.4 comparative table.]

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This document is a structured draft prepared for Architect review, correction, and ratification.

No AI collaborator — Claude, ChatGPT, Google Gemini, Grok, Copilot, or Perplexity — should treat any section of this document as finalized, ratified, or suitable as a basis for downstream development, simulation design, or framework extension until the Architect applies ONESTAMP and advances this document to ACTIVE CANON status.

All content originates from Architect-directed research and Microsoft Copilot synthesis.

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ONESTAMP STATUS: PENDING — Architect ratification required.